

The Effect of AI Writing on Governance: Evidence from a \\$3.5 Billion DAO

This paper investigates how the introduction of highly capable AI writing tools affected the informational value of a decentralized autonomous organization's governance forum. The authors find that before the release of Claude 3, forum activity predicted vote closeness, but this relationship disappeared post-release due to an unverifiability shock that made authentic signals indistinguishable from AI-generated text.

 Revision check BETA

This document is a resubmission of a previously reviewed version.

We checked your new version against the [previous feedback](#) and found: 0 fully addressed, 2 partially addressed, and 4 not addressed comments.

View revision check →

 Overall Feedback

Here are some observations and questions from a careful read of the document, focusing on the alignment between the empirical tests and the theoretical claims.

Structural break tests and timing

Readers will look closely at the evidence for an AI-induced slope break. The Chow procedure in Section 4.3 tests the stability of the full parameter vector rather than isolating the equality of the human-post slope, and Table 7 indicates that the direct slope change is +0.00222 with a standard error of 0.00181. Regarding the label-permutation exercise in Section 8.7, this approach removes temporal ordering, making it challenging to interpret as randomization inference for a calendar-time break.

There is also a question of timing alignment: the QLR test and the largest Chow statistic point to a break in November 2023, while Table 2 uses a March 2024 split because it yields a clearer pre-period relationship. Finally, the exclusion of 54 LTIPP/STIP proposals during this window introduces a compositional shift that category fixed effects cannot absorb. Demonstrating the persistence of the break within proposal types unaffected by the council-batching process would help distinguish an AI-induced slope change from concurrent calendar-time and composition effects.

Information aggregation versus latent proposal characteristics

The posts-margin estimand invites questions about whether it uniquely identifies information aggregation. A negative association between thread activity and vote closeness is equally consistent with latent controversy, proposal salience, complexity, or mobilization jointly generating more posts and closer votes, rather than necessarily showing that delegates learned from the forum.

In Sections 3.1–3.4, the document does not establish a cutoff restricting post counts to information available before Snapshot voting opened, leaving open the possibility that complete-thread counts include discussion generated after voting begins. Furthermore, the outcome captures token-weighted contestation while excluding abstentions and abstracting from turnout, passage thresholds, voting-power concentration, and vote rental. Without an auditable pre-vote information set and evidence that the forum measures add predictive content beyond pre-vote proposal characteristics, readers might interpret the result as changing descriptive predictability rather than a collapse of collective information aggregation.

The formal unverifiability mechanism

The connection between the formal unverifiability mechanism and the empirical results requires further development. In Proposition 1, expected posting is proportional to $p(1 - p)$ by assertion rather than being derived from specified contributor payoffs, and the representative-delegate setup does not clearly map to the observed token-weighted vote margin. More fundamentally, if latent proposal division drives both posting and voting, that covariance need not disappear merely because delegates stop reading posts.

The readership evidence in Section 7.2 presents a puzzle here: Discourse reads reflect session openings rather than identified delegate attention, and total user-post interactions appear to have increased by roughly 40% after the break even as reads per post declined. Finally, while heterogeneous thresholds are mentioned in a footnote, incorporating them into the model or linking them to delegate-level exposure would help resolve unverifiability from alternative channels like bounded attention, participation concentration, or post-load effects.

Measurement and sample construction

The audibility of the core AI measurement and sample construction will be a priority for replication. The Pangram classifier has not yet been validated on labeled Arbitrum prose, hybrid or AI-assisted writing, relevant model generations, or period-specific styles. Because detector error could attenuate estimates if it varies by period, model, length, or author type, bimodality and threshold invariance alone do not establish construct validity. This context complicates the human-post regressor and the central claim that realized AI volume was only 6–8% at the break.

Additionally, the sample counts exhibit internal variation: Section 2.5 notes 152 restricted proposals, Section 3.4 identifies 102 matched pairs, Section 8.4 discusses 24 unmatched proposals, Figure 8 refers to 292 matched threads, and Table 6 reports subsamples of 78 and 79 observations despite the 102-proposal main sample. An in-domain detector audit alongside a proposal-thread crosswalk to reconcile matches, exclusions, timestamps, scoring passes, and sample counts would secure the foundations of these central results.

Welfare and surplus calculations

The welfare calculations in Section 5.5 rely on a specific interpretation of bounds. The assumption that delegates' unobserved contamination belief π must exceed the realized AI share λ at the break does not follow from the data or the model, making $\pi^* \geq 0.082$ a maintained assumption rather than an identified bound.

Similarly, equating a post's information value with its 237-word writing cost functions as a normalization, and treating 62 self-linked active delegates as readers is unsupported by the readership measure. Consequently, the resulting claims—at least 858 words per delegate, 43,000 words per proposal, and complete destruction of surplus—compound these unverified assumptions about beliefs, valuation, readership, and the conversion of predictive correlation into decision quality. Framing these quantities as sensitivity calculations rather than empirical lower bounds would align the presentation with the available evidence.

Detailed Feedback

#1 Model cannot generate interior vote margins

 *A representative delegate (the decision-maker) observes all forum posts and votes For or Against the proposal; the proposal passes if the weighted vote share For exceeds threshold $\tau = \frac{1}{2}$. The delegate's payoff is +1 for a correct vote (For on G , Against on B) and -1 for an incorrect vote. Voting is costless. The delegate's information comes entirely from the forum.*

As written, the model cannot generate the interior vote margins used in the empirical analysis. A single representative delegate makes one binary choice, so the realized For share is either zero or one and the margin is always one. Interpreting the delegate as a population of identical voters does not help because they share the same prior, public information, and payoff. The later continuous object $\Phi(n_h, n_l; q)$ is therefore not derived from the setup.

#2 Section 5.2 does not derive the posts-margin result

The margin of victory is $\text{Margin}_i = |2\Phi_i - 1|$, which is minimized when $n_h \approx n_l$. Community division (p near $\frac{1}{2}$) produces roughly equal numbers of pro- and con-posts, so total volume is high and margins are low. Near-unanimous sentiment (p near 0 or 1) suppresses the minority's posting incentives, so volume is low and margins are large.

 Proposition 1 (Signal Content). In the no-AI equilibrium, forum post count is negatively correlated with the ex-post margin of victory: $\text{Cov}(\text{Posts}_i, \text{Margin}_i) < 0$.

Sketch. For proposal i with prior p_i , total posts $E[n_i] \propto p_i(1 - p_i)$, maximized at $p_i = \frac{1}{2}$. Expected margin $E[\text{Margin}_i] \propto |2p_i - 1|$, minimized at $p_i = \frac{1}{2}$. These are negatively correlated across proposals, since both are monotone functions of $|p_i - \frac{1}{2}|$ in opposite directions.

Proposition 1 does not follow from the stated primitives. Section 5.1 defines p_i as the prior probability that proposal quality is good, not as community support; contributor payoffs and pivotality are unspecified; and neither $E[n_i] \propto p_i(1 - p_i)$ nor the mapping from messages to the token-weighted vote margin is derived. The model does not specify contributor payoffs from the proposal outcome, influencing the delegate, signaling participation, or posting itself; as written, not posting strictly dominates. With a nonatomic continuum, an individual contributor has no specified pivotal influence. The model does not establish that posting is maximized at an even division, and neither two-sided truthful posting nor posting in proportion to support follows from the primitives. The asserted cross-proposal covariance therefore remains an assumption rather than an equilibrium result.

#3 Section 5.4 welfare loss is not implied by the model

Social welfare equals the expected correctness of the vote minus participation costs:

$$W = p \Pr(\text{pass} \mid \theta = G) + (1 - p) \Pr(\text{reject} \mid \theta = B) - c_p \cdot E[n].$$

In the no-AI equilibrium, the forum improves on the prior at social cost $c_p E[n]$. After the AI shock (for $\pi > \pi^*$), delegates vote at the prior: $W^{AI} \approx \frac{1}{2}$ when $p = \frac{1}{2}$, which is no better than a coin flip. The

 welfare loss is:

$$W^* - W^{AI} = 4p(1 - p) \left(q - \frac{1}{2} \right)^2 > 0.$$

This is maximized when $p = \frac{1}{2}$ (the most contested governance decisions) and when q is large (signals are informative). AI destroys the most welfare precisely where good governance matters most.

The welfare-loss formula $W^* - W^{AI} = 4p(1 - p)(q - 1/2)^2$ is not derived from the stated welfare function. The value of forum information depends on the number and selection of signals and the decision rule, while net welfare also depends on posting and potentially reading costs; prior-only optimal accuracy is $\max\{p, 1 - p\}$. Accordingly, neither the displayed formula nor its strict positivity follows from the model.

#4 AI-detection validity is not established in Section 3.2

Not all AI detectors perform equally well: Imas and Jaberian (2024) audit a range of commercial AI-text detectors and find that most fail to reliably distinguish AI-generated from human-written text, while Pangram achieves materially higher accuracy. Accurate detection is feasible, but requires a paid commercial API; free or open-source detectors are not reliable substitutes.³ Pangram analyzes the linguistic properties of a text passage and returns three complementary scores: fraction_ai (estimated proportion of content generated by a large language model), fraction_ai_assisted (proportion that is human-written but AI-edited), and fraction_human (proportion that is unassisted human writing). We focus on fraction_ai as our primary measure, applying a threshold of 0.70 to classify posts as AI-generated.

The AI measure lacks corpus-specific validation. Bimodality and invariance across thresholds establish classification stability, not accuracy; without a ground-truth check on DAO governance posts, differential errors across time, writing style, or delegate type could affect the human-post counts, AI shares, and delegate-level estimates. The treatment of fraction_ai_assisted also requires an estimand-based justification, because posts below the fraction_ai cutoff enter the human category even when their assistance score may be high.

#5 Chow test does not isolate the slope change in multiple sections

Columns (2) and (3) split the sample at the Claude 3 release date (March 4, 2024). In the pre-shock period (column 2, $n = 29$ proposals), the slope is $\hat{\beta} = -0.0026$ ($t = -2.30, p = 0.030$), corresponding to a Pearson correlation of $r = -0.33$. In the post-shock period (column 3, $n = 73$ proposals), the slope is $\hat{\beta} = -0.0006$ ($t = -0.92$), with $r = -0.10$. The null hypothesis of equal slopes across periods is rejected by the Chow test: $F = 8.38, p = 0.0004$.

The reported Chow test appears to test joint stability of the intercept and slope, not equality of the posts coefficient alone. This issue appears in the Abstract, Section 4.3, Section 6.1, Section 6.4, and Section 8.7 (Table 7). The QLR discussion uses $k = 2$, corresponding to the bivariate model rather than equation (1) with controls. Accordingly, its $p = 0.0004$ cannot be attributed specifically to a change in β ; moreover, Table 7's direct interaction estimate, $\Delta\beta = 0.00222$ with standard error 0.00181, does not reject slope equality at conventional levels. The controlled Chow test ($F = 3.35$) in Table 7 tests joint stability over multiple coefficients, potentially including the markedly changing turnout coefficient, so it shouldn't be attributed to the human-post slope alone. The text and table notes should distinguish omnibus parameter-stability tests from a one-restriction slope-change test, and the evidence for joint regression instability should therefore be distinguished from slope-specific evidence.

#6 Contradictory sample sizes for matched proposals

The matching procedure yields 102 matched proposal-thread pairs. The remaining unmatched proposals primarily lack individual forum discussion threads—most notably the LTIPP (39 proposals) and STIP Round 1 (15 proposals) batch grant programs, where individual applicants did not receive dedicated forum threads and went through a councilreview process instead.

The proposal counts are irreconcilable across the paper. Section 2.5 says the main analysis uses 152 proposals with the specific restriction of at least five human-authored forum posts, Section 3.4 says all main tests use 102 matched pairs, and Section 8.4 refers to 24 unmatched proposals. These figures do not fit the stated universe of 306 binary proposals or the 54 LTIPP/STIP examples. The sample-construction sequence does not explain how the filters reduce 306 proposals to these intermediate and final counts, so the observation flow and the population covered by the matching-bias analysis are not identifiable.

#7 Contradiction regarding Snapshot vote bindingness and data completeness

In the second stage, the proposer submits a "temperature check" to Snapshot, an off-chain voting platform that records token-weighted votes without requiring on-chain gas expenditure. Temperature checks are non-binding straw polls that provide a low-cost signal of likely vote outcomes. Proposals with clear temperature-check failures are typically withdrawn before proceeding.



In the third stage, passed temperature checks proceed to a binding on-chain vote via Tally, which records votes on the Arbitrum blockchain. A minimum quorum of 5% of circulating supply must participate for the vote to be valid, and the proposal must receive a majority (or supermajority, for constitutional proposals) of votes cast. Passed proposals are implemented automatically by the DAO's smart contract governor after a mandatory timelock delay (typically 14 days for non-constitutional and 21 days for constitutional proposals).

For our empirical analysis, we focus on Snapshot votes, which provide the best combination of coverage (both binding and informational votes are recorded)

The proposal lifecycle describes Snapshot temperature checks as non-binding and Tally votes as binding, yet the data discussion says Snapshot records both binding and informational votes. The abstract's characterization of the measured outcome as uniformly binding and of both datasets as complete overstates the design. The forum corpus excludes several categories and the estimating sample consists of 102 matched proposal-thread pairs. The later exclusion of 100 temperature checks does not identify what makes the remaining Snapshot ballots binding or how they relate to Tally. Without that mapping, the primary outcome cannot be interpreted confidently as either a preliminary sentiment measure or a final governance decision.

#8 Section 5.3 assumes the expectation jump that drives the break

The shock is summarized by two objects—a realized quantity and an expectation:



- $\lambda \in [0, 1]$: the realized fraction of posts in a thread that are AI-generated—a backward-looking measure of the contamination that has actually accumulated.
- $\pi \in [0, 1]$: the delegate's expectation that the next post read is AI-generated—a forward-looking object. In the long run π converges to λ , but at a capability shock π can jump ahead of λ : a delegate who knows a capable model now exists expects contamination to rise before it visibly has.

These two objects drive two distinct mechanisms. Whether the forum's signal degrades gradually (with realized volume) or discretely (with expectation) is an empirical question, and the two mechanisms make opposite timing predictions—which we exploit in Section 6.

The threshold result is conditional on an exogenous belief π , but the model does not derive why a model release makes π jump ahead of realized contamination λ or cross π^* . The timing evidence is consistent with anticipatory discounting, but without an adoption or belief-formation process it does not establish the paper's stronger claim that capability availability itself generates the expectation shock.

#9 Timing evidence does not by itself falsify volume

This explanation is falsified by the timing of the volume surge. Panel A of Figure 6 shows the monthly count of AI-generated posts ($\text{Pangram fraction_ai} > 0.70$). AI posts are detectable from the start of the sample—a small number of AI-assisted posts predate any large-scale adoption—but at single-digit volume: approximately 8% of forum posts at the QLR break date (November 25, 2023) and approximately 6% at the Claude 3 break date (March 4, 2024), dipping below 3% in between. The largescale volume surge arrives later: the AI share rises to roughly 10–15% through mid-2024 and above 20% only in late 2024, approximately nine to twelve months after the signal collapsed. We are not claiming the breakdown happens before any AI posts exist; we are claiming that the structural break precedes the surge—the point at which AI volume is large enough to mechanically dilute the human signal. A mechanism that requires large realized contamination cannot explain a collapse that precedes it.

The timing evidence weighs against a large-scale flooding account, but does not by itself falsify every realized-volume mechanism. The paper does not derive how much attenuation a 6–8% AI share should cause or analyze its proposal-level concentration, and measured contamination was already approximately 6–8% at the candidate break dates. Establishing that this share was mechanically insufficient would require a quantitative attenuation or dose-response analysis. Moreover, mechanically adding detected AI posts directly contaminates total-post counts, not the human-post count used in the main break analysis; effects on that variable require additional channels such as detector error or endogenous changes in human participation. The human-post specification is suggestive evidence against simple count dilution but does not exclude volume-induced changes in reading or human participation.

#10 Vesting-placebo counts exceed the main sample

Table 6 reports the results. The structural break is statistically significant in both subsamples. Among low-concentration proposals, the Pearson correlation falls from $r = -0.607$ pre-shock to $r = -0.149$ post-shock ($\text{Chow } F = 3.28, p = 0.043$). Among high-concentration proposals, it falls from $r = -0.233$ to $r = +0.097$ ($\text{Chow } F = 3.79, p = 0.027$). If the vesting cliff were the cause-by empowering whale voters to push margins wider—the break should appear only in high-concentration proposals. It does not. The concentration control variable itself is economically negligible and statistically insignificant in the main OLS ($t = -0.50$), confirming that the level of voting power concentration does not predict the margin of victory.

Table 6 appears inconsistent with the stated 102-proposal sample. Its concentration groups contain 78 and 79 observations, totaling 157, while its pre/post totals are 46 and 111 rather than 29 and 73. The table's full-sample correlations and Chow statistic also differ from the main results. The sample and specification underlying this placebo test therefore cannot be identified from the paper, so the reported table does not presently establish that the vesting cliff has been ruled out.

#11 Contradictory claims about cross-DAO data

Figure A7 shows that five DeFi DAOs with distinct governance structures—Aave, ENS, Compound, Gitcoin, and Optimism—all exhibit comparable participation declines over the same November 2023–March 2024 window

The cross-DAO evidence is internally inconsistent and therefore does not presently substantiate this identification claim. The text attributes Figure A7 to Aave, ENS, Compound, Gitcoin, and Optimism, while the Figure A7 caption describes only Arbitrum; Figure A6 separately says Optimism and Uniswap are excluded yet its caption lists them. The document also names different DAO samples elsewhere, leaving the analyzed comparison set and data source indeterminate.

#12 The 8% threshold calibration is not identified

Calibrating the threshold. The model implies a critical contamination rate $\pi^ = 1 - c_{\text{read}} / IH$ below which delegates read and above which they exit. The QLR break occurs when realized AI posts are approximately 8% of forum volume; because the delegate's expected contamination π runs ahead of this realized share, the break implies $\pi^* \geq 0.08$ (Section 5.5). The threshold condition then gives $c_{\text{read}} / IH = 1 - \pi^* \leq 0.92$: reading costs absorb at most 92% of the informational value of a perfectly reliable post. This is economically plausible in a setting where (i) each individual post contributes modestly to aggregate signal (the forum aggregates across hundreds of posts per proposal, so marginal informational value per post is small), (ii) careful reading of a governance post requires domain-specific knowledge and real attention, and (iii) a delegate cannot determine a post's type without first reading it, so expected value must exceed cost before type is revealed. Even a 5% probability of encountering a fully uninformative AI post is sufficient to erode the thin net return to deliberate forum reading. This calibration is consistent with the QLR break date and grounds the mechanism claim in observable data rather than leaving it asserted.*

The bound $\pi^* \geq 0.08$ is conditional rather than identified by the observed break and AI share alone: it requires the unobserved belief restriction $\pi \geq \lambda_{\text{break}}$. While it is economically plausible that forward-looking beliefs π exceed the historical realized average λ_{break} during a period of rising technology adoption, it should be explicitly identified as a substantive assumption about expectation dynamics rather than an algebraic certainty or 'fact'. Consequently, the implied cost bound and downstream welfare calibration are also conditional on expected contamination running ahead of realized contamination. A 5% contamination probability does reduce expected reading value, but the stated bound does not show that it triggers exit.

#13 Figure 3 does not display the stated taxonomy

Figure 3 shows the distribution of proposals by category and half-year period. Ecosystem program proposals are concentrated in the first half of 2024, which coincides with both the ARB vesting cliff and the Claude 3 release. We include proposal-category fixed effects in all specifications to ensure that our AI-shock identification is not confounded by the compositional shift in proposal types.

Figure 3: Proposal Volume by Economic Purpose. Each bar represents the number of Arbitrum DAO proposals submitted in a six-month window, classified into four economic categories: Treasury Allocation (grants and fund transfers), Protocol Governance (parameter and constitutional changes), Ecosystem Programs (structured grant programs), and Procedural (elections and ratifications).

Figure 3 appears inconsistent with its caption and the surrounding analysis. The plotted legend uses three categories—Treasury Spend, Rules Change, and Institution Building—rather than the stated four-category taxonomy; the displayed half-year totals sum to 400 rather than any identified analytical universe; and Panel (b) plots average For share despite being titled as vote margin. Consequently, the figure does not substantiate the claimed H1 2024 category shift or identify the categories and outcome used in the regressions.

#14 Panel design does not separate causal mechanisms

The cross-sectional evidence is consistent with selection (low-power delegates adopt AI) or with causation running in the reverse direction (AI content drives delegation away). To separate these mechanisms, we estimate a within-delegate panel specification:

$$\Delta \log \mathit{VP}_{it} = \alpha_i + \beta_1 \mathit{AI_frac}_{it} + \beta_2 \log \mathit{Posts}_{it} + \beta_3 \log \mathit{VP}_{it-1} + \varepsilon_{it}$$

where VP_{it} is delegate i 's voting power in 30-day period t , $\mathit{AI_frac}_{it}$ is their AI content fraction in that period, Posts_{it} is their log post count, α_i is a delegate fixed effect, and the dependent variable is the log change in voting power (approximating percentage change). Standard errors are clustered at the delegate level.

The fixed-effects panel does not separate selection from reverse causation. AI fraction and voting-power growth are measured contemporaneously within each 30-day period, and the specification supplies no exogenous variation or predetermined AI-use measure. It therefore establishes a conditional within-delegate association, not that AI content drives voting-power changes, attracts marginal delegators, or displaces posting volume as the causal factor.

#15 Event exposure makes distraction first stages endogenous

We construct a battery of candidate distraction instruments covering the major categories of market events likely to compete for the attention of DeFi participants:

Token airdrops. Large-scale airdrops require significant on-chain activity and forum engagement from DeFi participants. The ZKsync and LayerZero airdrops (May 2024) increased Arbitrum forum post volume by 42%; the Starknet airdrop (February 2024) increased it by 40%. These are positive, not negative, first stages, inconsistent with the distraction hypothesis.

Crypto market crashes. The Bank of Japan rate shock (August 2024), which triggered a 30% decline in ARB price within 72 hours, increased forum post volume in the surrounding two-week window ($\hat{\beta} = +0.983, t = 1.83, p < 0.10$).

For the event-window distraction tests, exposure is defined by whether a thread received at least one post during the event window, while the outcome is the thread's post count. This makes exposure partly dependent on realized posting and mechanically selects longer or more active threads into the exposed group. Exposed-but-silent threads are assigned to the control group, while treated observations are selected for positive activity. The resulting positive coefficients do not identify event-induced increases in posting and cannot, without an exposure definition independent of forum activity, rule out a negative distraction effect. This concern applies specifically to tests using that activity-based exposure rule, not automatically to the monthly volatility or concurrent-proposal specifications.

#16 Figures A3–A8 are paired with unrelated descriptions

Figure A5: Constitutional Amendment Lifecycle – Forum Discussion → DAO Vote → Outcome Arbitrum DAO, 2023–2026 (15 passed, 0 failed; 2 reduced participation barriers ↓)

Figure A5: Constitutional Amendment Lifecycle. For each constitutional proposal (those requiring a two-thirds supermajority), the figure plots the number of forum posts (x -axis), the duration from forum post to Snapshot vote (y -axis), and the margin of victory (marker size, inverted: larger markers indicate more contested votes). Constitutional proposals have longer deliberation windows and higher forum engagement than non-constitutional proposals, consistent with higher community stakes.

Figure A6: Cross-DAO Governance Benchmarking Voting Power Concentration, Participation, Passage Rate, and Minimum Control – Five DeFi DAOs on Snapshot, Jan 2023–Apr 2026 (Optimism and Uniswap use on-chain governance not accessible via Snapshot API and are excluded)

Figure A6: Cross-DAO Governance Benchmarking. Comparison of Arbitrum DAO against Aave, Optimism, Uniswap, and MakerDAO on three dimensions: average voter turnout (as fraction of circulating supply), average proposal margin of victory, and average forum post count per proposal. Arbitrum ranks in the middle tier on turnout and forum engagement. Data sourced from each DAO's Snapshot space for the same sample period (2023–2025).



Figure A7: Participation Breadth Over Time – Unique Voters per Proposal Five DeFi DAOs on Snapshot, Jan 2023 – Apr 2026 | Each dot = median voter count for proposals in that month

Figure A8: Panel Regression – AI Posting and Delegation Flows in Arbitrum DAO $N = 5354$ delegate $\times$ proposal observations | 52 delegates

Figure A7: Participation Breadth Over Time. Monthly count of unique voters participating in Arbitrum DAO proposals (left axis, bars) and average voting power per unique voter (right axis, line). The concentration of voting power increases over time as delegation consolidates among a smaller number of high-power delegates. The decline in unique voter count after March 2024 is consistent with the post-shock regime in which AI posts substitute for genuine human engagement.

Figure A8: Delegate Within-FE Panel Results. Coefficient plots for the panel specification in equation (4). Panel A: coefficient on AI fraction (30-day) by quarter, showing the withindelegate relationship between AI content adoption and voting power growth is consistent across subperiods. Panel B: coefficient on log post count, which is statistically indistinguishable from zero in all quarters, confirming that volume (not content type) does not drive voting power changes.

Figures A3–A8 do not match their prose descriptions. Figure A5 is a proposal timeline rather than the stated posts-by-duration scatter; Figure A6 uses different DAOs, variables, and four panels; Figure A7 is a five-DAO monthly voter series rather than an Arbitrum dual-axis plot; and Figure A8 is a rolling-correlation series rather than quarterly fixed-effects coefficient panels. This systematic mismatch also applies to Figures A3 and A4. The claims made in these captions are therefore not documented by the displayed graphics.

#17 Main specification is not the reported main regression

Our primary estimating equation is:

$$\text{Margin}_i = \alpha + \beta \text{HumanPosts}_i + \gamma X_i + \varepsilon_i$$

where Margin_i is the margin of victory for proposal i (defined in Section 3); HumanPosts_i is the count of human-authored posts in proposal i 's forum thread (posts with $\text{fraction_ai} \leq 0.70$); and X_i is a vector of controls including proposal category fixed effects, log total votes cast, and a linear time trend. Standard errors are heteroskedasticity-robust.

Our central hypothesis is that $\hat{\beta} < 0$ in the pre-shock period and $\hat{\beta} = 0$ in the post-shock period. We estimate equation (1) separately on the pre-Claude 3 subsample (proposals with Snapshot creation date before March 4, 2024) and the post-Claude 3 subsample, and test the null hypothesis $\beta^{\text{pre}} = \beta^{\text{post}}$ using a Chow (1960) test.

The declared primary specification includes category fixed effects, log votes, and a linear time trend, but Table 2 and the headline Chow statistic use an intercept-only bivariate model. Table 7 adds category effects and log votes but still omits the stated trend. The specification underlying each principal estimate and break test therefore needs to be aligned with equation (1), particularly because the controlled and uncontrolled Chow statistics differ materially.

#18 Word-equivalent surplus rests on an imposed ratio

The bound $\pi^* \geq 0.082$ implies $c_r = I_H (1 - \pi^*) \leq 0.918 I_H$. Normalizing $I_H = c_p = 237$ words (the information value of one genuine post equals the cost of writing one), this gives $c_r \leq 218$ words of reading and interpretation effort per post. The net reading value per delegate per post in the no-AI equilibrium is then bounded below:

$$I_H - c_r = \pi^* \cdot I_H \geq 0.082 \times 237 = 19.4 \text{ words per post.}$$

Equating I_H to c_p is presented as a 'normalization,' but it is a substantive assumption that fixes the ratio of two distinct economic quantities (the delegate's informational value of reading and the contributor's effort cost of writing) to exactly 1. A normalization sets the scale of units, but fixing $I_H = c_p$ establishes the exchange rate between effort and informational benefit. The conclusion that the forum is socially beneficial for $K \geq 13$ is mathematically driven by this assumption. Since the exact welfare bounds (19.4 words, 858 words) scale directly with this ratio, they are artifacts of this assumption. The paper later notes that delegates are largely non-pivotal and their private return to a correct vote is near zero, making it difficult to justify that a delegate's private informational value I_H is exactly equal to the physical effort cost c_p of writing 237 words. The tension between the assumed 1:1 ratio and the institutional reality of non-pivotal delegates should be addressed.

#19 Snapshot separation does not isolate information value

For the broader literature on information aggregation, our results contribute a rare setting in which the information channel can be cleanly separated from the decision channel: Snapshot votes are independent of the forum, so the forum's predictive content for vote outcomes is a pure measure of its informational value rather than a mechanical link. The collapse of this predictive content after AI shocks is perhaps the cleanest available evidence that generative AI degrades a specific form of collective intelligence.

Administrative separation between Snapshot and the forum rules out a mechanical accounting relationship, but it does not make the posts–margin association a pure measure of informational value. Because forum activity precedes voting, it may persuade, mobilize, coordinate, or select voters, while proposal characteristics may jointly affect posting and margins; these channels remain observationally combined with information aggregation.

#20 Conservative-bound claim in Section 3.5 is unsupported

The unmatched forum authors are disproportionately high-power participants (GFXlabs, DisruptionJoe, cp0x, CastleCapital) who do not disclose wallet addresses in public posts, likely for privacy reasons. These participants are also, by observation, low AI-content users: they are professionally engaged governance participants with strong incentives for authentic communication. Their exclusion therefore attenuates our estimated correlation between AI content and voting power toward zero—our estimate of $r = -0.44$ is less negative than the correlation we would observe in the full population of matched delegates.

The claim that $r = -0.44$ is a conservative bound does not follow from the observed linked sample. Most unmatched authors lack jointly observed voting power and AI use, and the professional status or incentives of several named delegates cannot determine the omitted observations' covariance or the full-population correlation. The delegate result can be described for the self-linked sample, but the direction of selection bias is not identified.

#21 The dilution calculation does not identify exit

This readership decline is not driven by post volume: the number of human posts on matched threads increased post-shock (from 2,650 to 5,898), so the per-post read count is not being diluted by fewer readers spread across more content—it reflects fewer readers per post on a forum that became objectively busier. To discipline this claim further, note that under a pure fixed-budget dilution story—in which aggregate reading engagement is constant and a doubling of posts mechanically halves reads per post—a 122% post increase would predict approximately a 55% decline in reads per post. The observed decline is only 36%, considerably smaller than the dilution benchmark. The implied aggregate reading engagement in fact grew post-shock: at the observed volumes, total user-post reading interactions increased by roughly 40% even as per-post reads fell. Aggregate reading grew; it simply failed to keep pace with the surge in post volume. This pattern is inconsistent with a pure fixed-budget story and is instead consistent with a mixed response: some delegates disengaged from the forum entirely (reducing the intensive margin of reading per post), while others—including new participants attracted by the busier forum—partially offset the loss.

The fixed-budget calculation does not identify delegate exit. The reported 40% aggregate-interaction estimate cannot be recovered by multiplying post counts by median reads per post, and the pre/post totals need comparable exposure periods. Even if total interactions rose, aggregate data cannot distinguish exit offset by entry from readership growth that merely lagged post growth. Complete exit is an extensive-margin, not an intensive-margin, change.

#22 Welfare question conflicts with the earlier calibration

For future research, three questions are most pressing. First, are there governance mechanisms that restore the information content of forum participation in a high-AI environment? Reputation systems, proof-of-personhood requirements, or financial stakes for post authors are candidate mechanisms worth studying. Second, do the patterns we document generalize to other DAOs? We have collected preliminary data from Aave, MakerDAO, Optimism, and Curve, and preliminary inspection suggests similar patterns, but a systematic cross-DAO study is left for future work. Third, what is the welfare consequence of the signal collapse? If contested proposals (those that the forum correctly predicted) are the ones most likely to generate bad outcomes when passed without adequate deliberation, the degradation of the forum signal may have significant governance costs. Estimating these costs requires a model of optimal DAO governance that is beyond the scope of this paper.

The conclusion is inconsistent about what welfare has been estimated. Sections 5.4–5.5 present model-based word-equivalent lower bounds of 858 words per delegate and roughly 43,000 words of social surplus per proposal, whereas this paragraph says the welfare consequence remains unestimated. The unresolved question is specifically the realized governance or treasury loss from worse decisions, not all welfare consequences of the signal collapse.

#23 Figure A10 conflicts with Table A2 and the conclusion

Figure A10 presents the first-stage results for each instrument category. Table A2 reports the corresponding regression estimates.

Note: Instruments with $n = 0$ treated proposals reflect no Arbitrum voting during the event window. Bitcoin halving (green, -sign) is confounded by the Arbitrum LTIPP grant batch ($n \approx 40$ small applications in April 2024). Full sample: $N=102$ contested proposals, 2023–2026.



Figure A10: Distraction Instrument Battery: First-Stage Effects on Forum Post Volume. Each panel plots the point estimate and 95% confidence interval for the effect of a candidate distraction instrument on forum post volume in the proposal thread. Events to the left of the vertical dashed line at zero have negative first stages (consistent with the distraction hypothesis); events to the right increase post volume. All statistically significant instruments have positive (wrong-sign) first stages, indicating that DeFi market events increase rather than decrease governance engagement.

Appendix D.2 appears internally inconsistent. Figure A10 reports a significant negative Bybit-hack estimate ($\hat{\beta} = -1.50, n = 2$), while Table A2 reports no overlap; the figure also gives the November 2023 cluster a negative estimate, whereas the table reports a positive one. Because the figure and table are presented as corresponding results, they do not presently support the stated conclusion that every statistically significant first stage has the wrong sign.

#24 Delegate evidence does not rule out power capture

Delegate-level evidence is consistent with this mechanism and rules out a power-capture story. Using a dataset linking forum usernames to Snapshot voting records for 60 delegates, we document a robust negative cross-sectional relationship between a delegate's average AI content fraction and their average delegated voting power: $r = -0.44 (p = 0.001, n = 51 \text{ delegates})$. High-AI delegates hold on average 71,000 ARB in delegated power; low-AI delegates average 1.9 million ARB—a gap of 27×. AI tools are adopted primarily by low-tier delegates seeking to maintain forum presence at lower cost, not by high-power insiders seeking to manufacture consensus. The signal degrades not because the powerful are gaming the forum, but because the possibility of doing so at zero cost became credible.

The delegate evidence rules out only the narrow claim that high average AI usage is concentrated among high-power delegates in the linked sample. Because wallet linkage is selected and the AI behavior of unmatched high-power delegates is not directly confirmed, the analysis cannot exclude strategic use by powerful actors or affiliated accounts. The broader inference depends on unverified assumptions about these 51 nonrandom, wallet-linked delegates and the unmatched high-power delegates' AI use. The cross-sectional association also does not identify lower-power delegates' motives, so the categorical power-capture and motivation claims exceed the design.

#25 Section 6.3 does not establish a complete mechanism

Ecosystem degradation: participation concentration. The content differences documented above are detectable in aggregate but not reliably in real time by individual delegates. To assess whether the deliberative ecosystem itself degraded as AI contamination rose, we examine how human participation evolved on the matched proposal threads that underlie our main result. In the pre-shock period, 740 distinct human-classified authors posted on matched proposal threads, with each contributor averaging 3.6 posts. Following the Claude 3 shock (March 2024 onward), the number of unique human participants fell to 496—a 33% decline—while posts per user more than tripled to 11.9. Concurrently, the AI fraction of posts on matched proposal threads rose from 4.7% to 11.3%, increasing the ratio of AI posts per human post from 0.05 to 0.13: a delegate scanning the forum now encounters 2.6× as many AI posts per human contribution as before the shock. These two forces together—sequential reading noise and participation concentration—provide a complete account of why the human-post → margin correlation collapses even though AI and human posts remain distinguishable in aggregate: the deliberative ecosystem itself narrowed, reducing both the expected return to reading any post and the informational content of the human posts that remain.

The claim that sequential reading noise and participation concentration provide a complete account of the correlation collapse is not established. The before-and-after participation totals pool unequal periods and 29 versus 73 proposals without normalizing author breadth by exposure, and the paper does not show that author concentration mediates the human-post coefficient or that repeated posts contain less information. These patterns support plausible complementary mechanisms, but not the stated causal sufficiency.

#26 Matching robustness uses an unexplained sample

We verify that the 24 unmatched proposals do not systematically differ from matched proposals along the dimensions relevant to our analysis. Unmatched proposals have slightly lower average vote participation (mean: 38 million ARB vs. 51 million ARB for matched proposals) but are not significantly different in margin of victory (mean: 0.63 vs. 0.65) or in the category distribution of proposal types. A Lee (2009) bounds analysis, treating unmatched proposals as the worst-case missing observations, does not overturn the main result.

The 24-proposal unmatched population is not reconciled with the paper's sample flow: the stated universes of 306 binary proposals and 152 proposals satisfying the deliberation restriction, followed by 102 matched pairs, do not identify a 24-observation unmatched set. Consequently, neither the matched-versus-unmatched comparison nor the stated Lee-bounds exercise is auditable without a complete accounting of the eligible population, sequential exclusions, and variables used in the bounds.

#27 Permutation labels do not give assumption-free inference

Bootstrap permutation Chow test. The Chow F -statistic of 8.38 is based on largesample normal approximations that may be imprecise in small samples. We therefore conduct a permutation test: under the null hypothesis of no structural break, we randomly reassign the pre/post labels of the 102 matched proposals, recompute the Chow F for each permutation, and repeat 10,000 times. Panel C of Figure 10 displays the resulting null distribution. The permutation p -value is 0.0009 (9 of 10,000 draws exceed the observed $F = 8.38$); the 95th and 99th percentiles of the null distribution are 3.64 and 5.32, respectively. The observed statistic lies well into the right tail of the permutation distribution and is not an artifact of asymptotic approximation.

The label-permutation exercise is not assumption-free small-sample inference for a break at a fixed calendar date. Its validity requires exchangeability of proposals across the pre/post partition, which is not established and may fail when proposal composition, covariates, or error distributions change over time. In addition, the classical Chow test has a finite-sample F reference distribution under normal, homoskedastic linear-model errors rather than relying inherently on a large-sample normal approximation. The permutation result is therefore diagnostic unless its exchangeability conditions are justified.

#28 Zero post-break value conflicts with readership evidence

At the tipping point, the delegate's net reading value falls from at least 858 words to exactly 0 in a single step, and at least 43,000 words of social surplus per proposal is destroyed. After the break, human contributors still write $\approx 8,858$ words per proposal (the post-period average), but delegates no longer read them—a pure transfer of human effort with zero social return. The welfare loss in Section 5.4 is thus not a continuous erosion but a cliff: a realized AI share as low as 8% marks the point at which the forum transitions from a public good worth at least 43,000 words per proposal to a 0-word-per-proposal waste.

The main text claims that after the break, the reading value falls to 'exactly 0', delegates 'no longer read them', and the forum becomes a '0-word-per-proposal waste.' This absolute language contradicts the empirical evidence presented in Section 7.2, which documents a 36% decline in median reads (from 88 to 56), not a drop to zero. While the theoretical representative-agent model predicts a discrete cliff, the empirical claims about the actual forum's collapse should be softened to align with the measured partial degradation.

#29 Contradictory timing in the reading decision rule

A delegate's decision to read post j is made before its content is observed: reading costs c_{read} are incurred before the post's type is revealed. This matters even when posts are partially distinguishable ex post. Our TF-IDF evidence (Section 6) confirms that AI and human posts differ in aggregate vocabulary; a delegate cannot determine which posts are human without first reading them.

The timing of the reading decision is internally inconsistent: the paper first conditions the decision on a post-specific posterior derived from textual features, but here says the decision and its cost precede observation of the post's content. The mechanism requires a defined sequence distinguishing any low-cost preliminary screening from costly reading and interpretation; otherwise the relevant pre-reading belief is the base rate π_j , not the feature-conditioned π_{jp} .

#30 Misapplication of Lee (2009) bounds to a slope coefficient

We verify that the 24 unmatched proposals do not systematically differ from matched proposals along the dimensions relevant to our analysis. Unmatched proposals have slightly lower average vote participation (mean: 38 million ARB vs. 51 million ARB for matched proposals) but are not significantly different in margin of victory (mean: 0.63 vs. 0.65) or in the category distribution of proposal types. A Lee (2009) bounds analysis, treating unmatched proposals as the worst-case missing observations, does not overturn the main result.

The Lee-bounds claim is not sufficiently specified for the paper's difference-in-slopes estimand. Standard Lee trimming bounds mean effects under monotone selection; they do not automatically bound a posts-margin slope or Chow statistic, particularly when unmatched proposals lack a well-defined human-post count. The paper needs to identify the bounded estimand, selection and monotonicity assumptions, trimming or imputation procedure, and treatment of the missing regressor before this can support the main result.

#31 Figure 6 does not show the stated break timing

Figure 6 presents the main causal figure. Panel A shows the evolution of monthly forum post volume, decomposed into human-written (Pangram AI fraction ≤ 0.70) and AI-generated (AI fraction > 0.70) posts, with AI model release dates marked by vertical dashed lines. Panel B plots the rolling 25-proposal Pearson correlation between human post count and margin of victory as a function of proposal date. The correlation is stably negative at approximately -0.4 through the pre-shock period, then rises to approximately zero following Claude 3. Panel C repeats the exercise using total post count (human plus AI), which shows a similar pattern with a faster deterioration, consistent with AI posts diluting the human signal. The three-panel figure is the primary evidence for our central result.

Figure 6 does not display a clean event-aligned break at Claude 3: the rolling correlation begins changing before the marker, reverses sharply afterward, and approaches zero only later in 2024. Because the rolling series has no observations around the November 2023 GPT-4 Turbo release, it also cannot visually corroborate the paper's preferred endogenous break timing. The figure can illustrate longer-run attenuation, but not the discrete timing attributed to it.

#32 Contradiction between text and Table 3 regarding signal recovery

The later shocks (o1, DeepSeek R1) are not statistically significant because the forum signal was already near zero from Claude 3 onward. All six candidate dates show the same directional break: the pre-period correlation is negative and the post-period correlation is near zero.

The claim that the signal remained near zero appears inconsistent with Table 3's DeepSeek R1 row, where the post-break correlation is $r = -0.38$ rather than near zero and is more negative than the pre-break value of -0.17 . This estimate may suggest later recovery, but with only 24 post-break proposals and an insignificant Chow test it does not establish recovery; the paper should nevertheless avoid describing every post-Claude 3 subsample as near zero.

#33 Mechanism claim in the conclusion exceeds identification

The mechanism is unverifiability, not manipulation. The signal collapses not because AI posts flood the forum—they were in the single digits (about 8%) of volume at the break—but because the credible possibility of indistinguishable AI content is sufficient to destroy delegates' incentive to condition their votes on forum activity. This is precisely the Akerlof (1970) mechanism: market exit precedes the accumulation of lemons because rational agents respond to updated priors, not realized averages. AI-generated posts carry zero information about vote outcomes (their correlation with margins is -0.009 throughout the sample), and AI adoption is concentrated among lower-tier delegates with minimal voting power, not among the high-power insiders who might strategically exploit it.

The conclusion states the unverifiability mechanism too categorically. The timing of the break, decline in reads per post, and delegate evidence are consistent with belief-driven discounting, but the analysis does not observe delegates' contamination beliefs or link delegate-level reading changes to votes. The evidence therefore supports unverifiability as the leading interpretation rather than uniquely identifying it.

#34 Figure A4 mixes incompatible treasury units and levels

Figure A4: Fiscal Evolution & Treasury Dynamics Arbitrum DAO – Quarterly ARB Approved by Governance, 2023 Q1–2026 Q1 (Layer 1: curated major programs; Layer 2: parsed Treasury Spend proposals; balance inferred from approved disbursements)

 *Figure A4: Fiscal Evolution and Treasury Dynamics. Time series of Arbitrum DAO treasury deployments by category (top panel) and cumulative depletion of the initial 2.75 billion ARB DAO treasury (bottom panel). Treasury values are converted to USD at the prevailing ARB price at the time of each disbursement. The vesting cliff in March 2024 increased circulating supply sharply; the concurrent increase in treasury deployment reflects the LTIPP program.*

Figure A4 uses incompatible units and baselines. Both panels are labeled in ARB, despite the caption stating that treasury values are converted to USD, and the balance series begins near 3.5 billion ARB rather than the stated initial 2.75 billion ARB. The figure therefore does not currently define a coherent treasury accounting identity.

#35 Snapshot sample counts are internally inconsistent

The raw pull yields 406 closed proposals. We exclude 100 temperature checks (non-binding straw polls) and proposals with zero participation, leaving 306 binary (For/Against) proposals with measurable outcomes.

 *Our primary outcome variable is the margin of victory: the absolute difference in vote share between the winning and losing choices, restricted to the For and Against categories (excluding Abstain). Formally, for a proposal i with For votes s_F and Against votes s_A :*

The Snapshot sample flow is internally inconsistent. The paper says that removing temperature checks and zero-participation proposals leaves 306 binary proposals, but then describes a subsequent exclusion of non-binary proposals without reporting a new count. It is also unclear whether the 100 observations are temperature checks alone or the union of both stated exclusions, so the analysis universe cannot be reconstructed from the reported numbers.

#36 Table 1 classifications exceed scored-post coverage

Table 1: Summary Statistics: Arbitrum Governance Forum

	Mean	Median	Std. Dev.	Min	Max
Posts per topic	15.9	7.0	28.3	1	412
Human posts per topic	13.2	5.0	24.1	1	389
AI posts per topic	2.7	0.0	8.4	0	143
AI fraction (post level)	0.11	0.00	0.27	0	1.00
Words per post	312	148	487	1	8,204
Panel B: By period					
		Pre-Claude 3	Post-Claude 3		Change
	Mean	Std. Dev.	Mean	Std. Dev.	
Monthly posts	387	201	512	298	+32.3%
Monthly AI posts	8	11	143	127	+1,688%
AI fraction (monthly)	0.02	0.03	0.28	0.18	+0.26

Notes: Panel A reports post-level and topic-level summary statistics for the full sample period (January 2023–March 2026). Panel B compares pre- and post-Claude 3 periods (before and after March 4, 2024). AI posts are those with fraction_ai > 0.70 as classified by the Pangram API. Human posts are all other scored posts.

Table 1 appears internally inconsistent about scoring coverage. Mean human and AI posts per topic sum to the mean total posts per topic, implying that virtually all 17,553 posts are classified, whereas Section 3.2 reports only 15,891 scored posts and Appendix A.4 excludes unscored posts from both categories. The reported total, scored, human, AI, and unscored post quantities therefore cannot all use the stated definitions and common topic denominator.

#37 Active-delegate count does not establish readership

If K delegates read the forum, the net social surplus per proposal is at least

$$K \cdot 858 - \bar{n} \cdot c_p = 858K - 10,480 \text{ words, where } \bar{n}$$

which is positive for $K \geq 13$. Arbitrum’s 62 identified active delegates ($K = 62$) imply net social benefits of at least $62 \times 858 - 10,480 \approx 43,000$ words per proposal—about four times the human writing effort invested in each thread—entirely from the public-good externality of the forum. Because 454 unique community members posted in pre-period proposal threads (and readership exceeds participation), $K \geq 62$ is itself a lower bound, so the 43,000-word figure compounds two conservative margins.

The calibration uses $K = 62$ to calculate a per-proposal social surplus of 43,000 words. However, 62 is the total number of identified active delegates across the entire sample. Substituting this aggregate count into a per-proposal equation assumes that all 62 delegates read every post of every single proposal thread. Using a corpus-wide metric for a per-proposal parameter likely inflates the resulting welfare estimates.

#38 Figure A1 does not locate the claimed threshold

Figure A1: Independent Evidence for the AI Indistinguishability Threshold. Share of interrogators who judged the AI to be human across three Turing-test studies: Jones and Bergen (2025b)'s earlier online 2-player game (GPT-4, best prompt 49.7%), their randomized, preregistered 2-player test (GPT-4, 54.0%), and Jones and Bergen (2025a)'s 3-player test (GPT-4.5, 73.0%). The dashed line marks chance (50%) and the dotted line the human baseline (66–67%); the shaded band is the November 2023–March 2024 capability-threshold window (GPT-4 Turbo to Claude 3), within which the frontier crosses from chance to above chance. Caveats: the three studies use different designs (online vs. controlled 2-player vs. 3-player) and the 2025 point requires prompting the model to adopt a humanlike persona; the connecting line indicates the frontier over time, not a single controlled panel. No benchmark measures indistinguishability of governance-forum prose specifically—these are the nearest available instruments.

The timing claim in Figure A1 is overstated. The graphic contains no direct measurement for GPT-4 Turbo or Claude 3 within the shaded interval; its apparent crossing is an interpolation between differently designed GPT-4 tests on either side of that interval. The evidence can illustrate a broad trend toward outputs becoming harder to distinguish, but it does not independently locate an indistinguishability threshold in November 2023–March 2024 or specifically for governance-forum prose.

#39 Abstract states sufficiency beyond the identification

Interestingly, we observe that the degradation of the forum's signal (the elimination of the correlation between posting volume and margin-of-victory) takes place at a time when relatively few posts are A.I. generated, relative to the end of the sample period where A.I. posting volume grows rapidly. We use a model of endogenous posting and voting to show that this may be a general phenomenon: the signal value of the forum can collapse to zero at a discrete tipping point with a very low volume of A.I. posts. Calibrating to the data, the forum generated net social benefits equivalent to saving at least 858 words of research effort per delegate per proposal—all destroyed at the tipping point. Delegates who rely on AI hold far less voting power than those who do not, ruling out power capture by insiders. Our results suggest that the mere existence of indistinguishable AI writing is sufficient to degrade the governance capabilities of economically important organizations, even if such AI is rarely used in equilibrium.

The claim that mere availability of indistinguishable AI writing is sufficient for degradation is stronger than the model and design establish. The model requires expected contamination to exceed a reading-cost threshold, while the empirical evidence does not directly observe those expectations and remains consistent with, rather than uniquely identifying, the proposed unverifiability mechanism.

#40 Matching threshold leaves the 102-pair sample unclear

Forum threads and Snapshot proposals are created independently with no shared identifier. We match them by fuzzy title similarity using the Ratcliff/Obershelp sequence-matching algorithm in Python's difflib library (SequenceMatcher class). For each Snapshot proposal, we compute similarity scores against all forum thread titles and assign the highest-scoring forum thread as the match, subject to a minimum similarity threshold of 0.55. We audit all matches with similarity between 0.55 and 0.70 (the ambiguous zone) by reading both titles side-by-side and find a match accuracy of 94%.

The final construction of the 102 matched pairs is ambiguous. The paper reports the yield after describing a 0.55 candidate threshold, then states that a one-to-one constraint and a 0.65 minimum are also imposed, without establishing whether 102 is the count after those filters or how failed manual audits were handled. Because a mismatched thread changes the measured post count and can cross the break date, the final matching rule and sample flow are substantively important.

#41 Mathematical inconsistency in Table 1 summary statistics

Posts per topic | 15.9 | 7.0 | 28.3 | 1 | 412 || Human posts per topic | 13.2 | 5.0 | 24.1 | 1 | 389 || AI posts per topic | 2.7 | 0.0 | 8.4 | 0 | 143 || AI fraction (post level) | 0.11 | 0.00 | 0.27 | 0 | 1.00 |

Table 1 appears mathematically inconsistent under its stated common-sample definitions. The topic-level means imply that AI-classified posts constitute roughly 17% of all posts; because each such post has a Pangram score above 0.70, their contribution alone would place the overall mean score above the reported 0.11, even allowing for the displayed rounding. The rows therefore appear to use different samples or contain a classification or aggregation error.

#42 Section 6.3 overstates what the null correlation shows

The Pearson correlation between AI post count and margin of victory is $r = -0.009$ ($p = 0.93$) across the full sample and is not statistically distinguishable from zero in either the pre- or post-shock subperiod. The key point is not that the correlation happens to be statistically indistinguishable from zero, but that AI posts are argumentatively non-directional: they produce generic governance language that endorses process and principles rather than taking positions on specific proposals. Their content does not reveal private information about θ_i , so they add volume without adding signal. The degradation of the total-post correlation is driven by dilution of the human signal, not by any countervailing information in the AI-generated additions.

Section 6.3 overstates the evidence: the near-zero correlation between AI-post count and absolute vote margin shows only that AI posting volume is unrelated to contestation, not that individual AI posts are non-directional or reveal no private information. The separately mentioned pass/fail null and the TF-IDF evidence support at most the weaker conclusion that AI-classified posts are generic and provide no demonstrated proposal-level predictive signal; indeed, the prevalence of support phrases and conditional endorsements shows that some are linguistically directional.

#43 Readership timing is not pinned to the release date

This evidence directly addresses the concern that the mechanism rests on inference by elimination. Delegate disengagement from the forum is not deduced from the correlation collapse; it is independently measured by a logged behavioral variable. The timing-declining reads beginning in November 2023, nineteen days after GPT-4 Turbo was released—is consistent with rational anticipation rather than documented contamination: delegates reduced forum engagement in response to the credible possibility of AI contamination, before the explicit governance discourse about AI post quality emerged (which we observe only from 2025 onward). This sequence is precisely what the unverifiability mechanism predicts: market exit precedes the accumulation of case-by-case evidence, because rational agents respond to priors, not realized averages.

The exact nineteen-day timing comes from the November 25 QLR break in the posts-margin relationship, not from Figure 8's monthly readership series. The readership data show only a small change from roughly 100 to 97 in November, with larger declines in later months, so they support at most a broad post-November decline rather than readership exit dated nineteen days after GPT-4 Turbo or an immediate response to that release.

#44 Proposal fixed effects cannot support this design

Third (exclusion restriction), the AI shock affects the margin of victory only through its effect on the informational content of the forum, not through any other channel. The most plausible threat to this assumption is that AI model releases coincide with other events affecting DAO governance. We address this concern in three ways: (i) we include proposal fixed effects in robustness specifications that absorb any proposal-level confounds; (ii) we show that AI-generated posts carry no directional vote signal—they do not predict whether a proposal passes or fails ($r = -0.009, p = 0.93$)—which is distinct from the claim that the correlation happens to equal zero; rather, AI posts are argumentatively neutral, echoing whichever side has more forum presence without revealing genuine opinion; and (iii) we test and reject a comprehensive battery of distraction instruments that might confound the before-after comparison (Section 8).

The claim that proposal fixed effects absorb proposal-level confounds is infeasible in the 102-observation proposal-level cross-section. Because each proposal appears once, proposal indicators would absorb all proposal-level variation, including HumanPosts. The reported robustness specifications instead use proposal-category fixed effects, which control for average category differences but not arbitrary proposal-specific confounding.

#45 ENS matching step in Appendix A.3 is incomplete

We focus on the first post in each delegate's introductory thread (identified by searching for threads titled "[Delegate Pitch]" or similar), where self-disclosure of wallet addresses is most common. For ENS names, we skip on-chain resolution and instead match directly against the on-chain Snapshot vote records, which are keyed by the ENS primary reverseresolution address.

Appendix A.3 does not identify the operative ENS-to-voter linkage. Snapshot vote records are described elsewhere as keyed by Ethereum address, whereas the extracted ENS item is a name; reverse resolution ordinarily maps an address to a name and does not convert a disclosed name into the voting address. Unless the procedure compared the disclosed name with a reverse-resolved name field attached to a Snapshot voter address, the stated direct match cannot produce the reported links. This leaves an important component of the delegate analyses insufficiently reproducible.

#46 Benchmarks do not locate the claimed capability threshold

There is direct benchmark evidence that these two releases were qualitatively different from their predecessors. On the Massive Multitask Language Understanding benchmark (MMLU; Hendrycks et al., 2021), which tests breadth of knowledge across 57 academic and professional domains, GPT-4 (March 2023) scored 86.4% versus 70.0% for GPT-3.5—a step change that closed most of the remaining gap to expert human performance (estimated at 89.8%). GPT-4 Turbo extended this to 87.3%, and Claude 3 Opus reached 86.8% (Anthropic, 2024). On MT-Bench, a benchmark designed to evaluate multi-turn instruction-following and writing quality, GPT-4 achieved a score of 8.99 out of 10 versus 7.94 for GPT-3.5 (Zheng et al., 2023). Most relevant for governance prose, evaluations of long-form argumentative and analytical writing—the genre of DAO forum posts—show a discrete jump precisely at GPT-4 Turbo, with evaluators unable to reliably distinguish outputs from expert human writers at threshold quality settings (OpenAI, 2023). Prior models (GPT-3.5, Claude 1, Claude 2) produced outputs that trained evaluators could identify with roughly 70–75% accuracy; GPT-4 Turbo and Claude 3 reduced this to near chance (Anthropic, 2024).

The displayed MMLU and MT-Bench comparisons do not themselves locate an indistinguishability threshold in November 2023–March 2024; their reported large contrast is between GPT-3.5 and GPT-4 in March 2023. The later timing therefore rests primarily on the asserted long-form indistinguishability results, whose connection to governance-forum prose is necessarily indirect, so the evidence supports the proposed threshold window more cautiously than the phrase “direct benchmark evidence” suggests.

#47 The reported correlation is not directional

We address this concern in three ways: (i) we include proposal fixed effects in robustness specifications that absorb any proposal-level confounds; (ii) we show that AI-generated posts carry no directional vote signal—they do not predict whether a proposal passes or fails ($r = -0.009$, $p = 0.93$)—which is distinct from the claim that the correlation happens to equal zero; rather, AI posts are argumentatively neutral, echoing whichever side has more forum presence without revealing genuine opinion; and (iii) we test and reject a comprehensive battery of distraction instruments that might confound the before-after comparison (Section 8).

The statistic $r = -0.009$ is the reported correlation between AI-post count and the nondirectional margin of victory, not the pass/fail outcome. It therefore does not establish the absence of a directional vote signal, and by itself it also cannot establish that AI posts are argumentatively neutral or reveal no genuine opinion. The paper mentions an analogous directional null elsewhere but does not report that statistic here.

#48 Post opens do not directly measure delegate reading

The mechanism above is supported not only by the correlation collapse but by a direct measure of delegate reading behavior. Discourse, the platform hosting the Arbitrum governance forum, records a reads count for each post: the number of distinct user sessions in which the post was opened. This is not a proxy—it is a logged count of how many users actually read the post. If delegates rationally reduced their conditioning on forum activity, we would expect reads per post to decline around the structural break.

The Discourse reads count records distinct sessions in which a post was opened, not identified delegates or verified careful reading. It is therefore an aggregate exposure or readership proxy; by itself, it cannot directly establish delegate-level disengagement or that delegates stopped conditioning their votes on forum information.

#49 Invalid interpretation of dynamic panel coefficients

The mean reversion coefficient is -0.047 per period, implying that voting power systematically declines toward a delegate's long-run level; the positive AI coefficient means only that Alactive periods exhibit slower decay, not genuine accumulation. A plausible interpretation is that the appearance of activity momentarily slows delegator attrition, but the effect is economically small relative to mean reversion and quickly dissipates

The dynamic-panel interpretation is not supported by the estimated coefficients. A negative coefficient on lagged log voting power implies conditional mean reversion, not unconditional decay, while the positive AI coefficient can generate positive predicted growth and turn predicted contraction into positive growth. The raw coefficients 0.055 and -0.047 also have differently scaled regressors and cannot be compared directly. Any mean-reversion effect depends on the delegate's distance from the conditional steady state, so it cannot be directly compared with the 5.5% full-range AI-fraction effect or said to "quickly dominate" it. If the model is interpreted as a stable adjustment equation and AI fraction undergoes a permanent exogenous increase from zero to one, the implied steady-state difference is $-0.055/(-0.047) \approx 1.17$ log points; however, that extrapolation is conditional on strong persistence and causal assumptions and should not itself be presented as an established tripling of voting power. The regression also does not establish that the association quickly dissipates or reflects eventual delegator penalties.

#50 Figure A3 caption conflicts with the vote-rental series

Figure A3: The Vote Rental Market – LobbyFi on Arbitrum DAO Delegated Voting Power and Governance Share of the LobbyFi Proxy, July 2024 – April 2026 (Each dot = one Snapshot proposal vote; 78 votes total, 0 after October 2025)

 *Figure A3: The Vote Rental Market: LobbyFi on Arbitrum DAO. Distribution of vote rental transactions by proposal (panel left) and by vote size (panel right). Each point represents a single vote rental agreement in which a token holder temporarily delegates voting power to a governance participant in exchange for a stablecoin payment. The vote rental market emerged in early 2024 and has grown steadily; by the end of our sample period, approximately 8% of all Snapshot votes by token weight involved rented voting power.*

Figure A3's caption does not match the displayed analysis. The panels are time series of the LobbyFi proxy's voting power and proposal-level share of voting power cast, not distributions of individual rental agreements. The graphic also shows no votes after October 2025, so it does not support steady growth through the sample or, without an additional aggregation, the stated 8% share of all Snapshot voting power.

#51 Market-efficiency analogy does not imply no direction

Antweiler and Frank (2004) find that Yahoo Finance message board posts predict return volatility and trading volume but carry minimal directional content (expected return predictability). The theoretical reason is the semi-strong form of market efficiency (Fama, 1970): forum posts are public information, and public information cannot systematically predict the average direction of asset returns—any such content would already be incorporated into prices. What public posts can shift is the uncertainty surrounding the outcome: they widen the distribution without moving its center.

Our governance setting mirrors this structure exactly. Arbitrum forum posts are publicly observable by all token holders, making them public information in the Fama (1970) sense. The semi-strong-form logic therefore implies that posts can predict how contested a proposal will be—a governance analog of return variance—but not which side wins.

The semi-strong-efficiency argument does not directly apply to a future DAO vote because there is no contemporaneous market price that incorporates public forum information and thereby removes predictable directional adjustment. Public forum posts can therefore predict which side will win as well as how contested the vote will be. The near-zero correlation between total post count and pass/fail is a descriptive empirical result—not a theoretical implication of the posts being public—and does not rule out directional information in the balance or content of pro- and con-posts.

#52 Figure 8 reports an incompatible matched sample

[^4] Figure 8: Forum Readership Decline on Matched Governance Proposal Threads. Monthly median reads per human-authored post on the 292 matched proposal threads used in the main analysis. Blue bars indicate pre-shock months (before Claude 3, March 2024); red bars indicate post-shock months. Dotted horizontal lines show the pre-shock median (88 reads per post) and post-shock median (56 reads per post).

Figure 8 is said to use 292 matched proposal threads from the main analysis, whereas Sections 3.4 and 6.1 repeatedly define that sample as 102 matched proposal-thread pairs. The readership sample is therefore internally inconsistent and must be reconciled before the figure can be interpreted as evidence from the threads underlying the main result.

#53 LTIPP/STIP bias direction appears reversed

The LTIPP/STIP batching, if anything, would strengthen the posts-margin correlation in the pre-period: competitive grant allocation concentrates genuine deliberation on contested funding decisions, which should tighten the link between forum activity and vote contestation. An AI-unrelated LTIPP/STIP effect therefore biases against finding a collapse. The vesting cliff, tested directly in Section 8.5, is ruled out by timing: the QLR endogenous break falls in November 2023, four months before the cliff, and the structural break is present in both high- and low-voting-power-concentration proposal subsamples. Vote-rental growth would attenuate the posts-margin correlation throughout the sample uniformly rather than generating the sharp structural break we identify.

The stated direction of the LTIPP/STIP bias is reversed under the paragraph's own hypothetical. If batching makes the pre-period posts-margin relationship more negative while leaving the post-period relationship unchanged, it enlarges the pre/post difference and biases toward finding a collapse, not against it. The effect's actual direction is additionally uncertain because many batch proposals were unmatched and excluded.

#54 Candidate-date claims conflict with Table 3

The later shocks (α_1 , DeepSeek R1) are not statistically significant because the forum signal was already near zero from Claude 3 onward. All six candidate dates show the same directional break: the pre-period correlation is negative and the post-period correlation is near zero.

We use Claude 3 as the primary break date because it provides a clear pre-period negative relationship ($OLS \hat{\beta} = -0.0026, p = 0.030, n = 29$) and no post-period relationship ($r = -0.10, p = 0.39$). All results are robust to using GPT-4 Turbo or any other candidate date; see Table 3.

Table 3: Chow Test Battery: Structural Breaks at AI Model Release Dates

The candidate-date conclusions appear broader than the results in Table 3 support. Claude 2 changes from $r = -0.17$ to -0.22 , α_1 from -0.20 to -0.18 , and DeepSeek R1 from -0.17 to -0.38 ; these rows do not all show movement toward zero, and their Chow tests are insignificant. The evidence supports a cluster of breaks around GPT-4 Turbo through GPT-4o, not robustness to every candidate date.

#55 Mathematical impossibility in token distribution

On March 16, 2023 (the "Token Generation Event," or TGE), the Arbitrum Foundation distributed 10 billion ARB governance tokens according to the following schedule: (i) 11.62% to eligible Ethereum users via a public airdrop; (ii) 1.13% to eligible DAOs building on Arbitrum; (iii) 42.78% to Offchain Labs team members and advisors, subject to a four-year vesting schedule with a one-year cliff; (iv) 17.53% to early investors, subject to the same vesting schedule; and (v) 27.70% to the Arbitrum DAO treasury, available for governance allocation.

The allocation percentages sum to 100.76%, so at least one reported share is incorrect. In addition, if the stated four-year team-and-investor vesting schedules released 25% at the one-year cliff, the reported 60.31% combined allocation would imply approximately 1.51 billion ARB unlocking, not the stated 1.1 billion. Furthermore, the 27.70% treasury share corresponds to 2.77 billion ARB, whereas the appendix refers to an initial 2.75 billion ARB treasury, a difference that affects the stated division between insider and DAO holdings. The internal evidence does not by itself establish that the team and treasury figures were simply swapped, but it does show that the allocation and vesting quantities require reconciliation.

#56 Concentration metric does not establish contestation

Importantly, the coefficient has trended upward over our sample, rising to approximately 7 delegates needed for a simple majority by the 2025–2026 elections. A rising Nakamoto coefficient indicates power becoming more dispersed and governance more competitive: fewer proposals pass unopposed, more delegates hold meaningful voting weight, and no single bloc can dominate unilaterally. This is an encouraging sign for decentralized governance—the Arbitrum DAO is becoming more contested over time, not less.

2.4 The Vote Rental Market

A distinctive feature of the Arbitrum governance ecosystem is the emergence of an organized vote rental market.

The rising Nakamoto coefficient supports greater dispersion of measured voting power, but it does not by itself establish that fewer proposals pass unopposed, that governance has become more contested, or that no coordinated bloc can dominate. Those are distinct outcome and coalition claims requiring evidence beyond the concentration statistic in Figure 2.

#57 Unsupported visual claim about pre-shock trend

Panel B plots the rolling 25-proposal Pearson correlation between human post count and margin of victory as a function of proposal date. The correlation is stably negative at approximately -0.4 through the pre-shock period, then rises to approximately zero following Claude 3.

Because only 29 proposals predate Claude 3 and the rolling window contains 25 proposals, Panel B provides at most five pre-break endpoints, all based on heavily overlapping windows. Although those windows incorporate earlier proposals, the figure cannot establish that the correlation was stable throughout the January 2023–February 2024 pre-shock period.

#58 Delegate post-count cutoff conflicts in Appendix A.4

Human and AI post counts. For each forum topic linked to a Snapshot proposal, we count posts with $\text{fraction_ai} \leq 0.70$ as human posts and posts with $\text{fraction_ai} > 0.70$ as AI posts. Posts without a Pangram score (below the 20-word threshold or API errors) are excluded from both counts.

Delegate AI fraction. For each delegate, average AI fraction is the mean fraction_ai across all their scored posts in the sample period. Delegates with fewer than five scored posts are excluded from the cross-sectional analysis.

The delegate cross-section's inclusion rule is internally ambiguous. Section 3.5 traces the reported 51 observations through 52 delegates with at least 10 total posts, whereas Appendix A.4 states that delegates need at least five scored posts. Total and scored posts are different because sub-20-word posts are unscored, and the paper does not establish whether both filters were applied or which rule generated the reported sample.

#59 Threshold timing exceeds the benchmark evidence

Second, controlled Turing tests: in a randomized, preregistered two-party test, Jones and Bergen (2025b) found GPT-4 judged to be human 54% of the time—versus 66–67% for real humans and 22% for a 1960s eliza baseline—up from 49.7% for the best GPT-4 prompt in an earlier online study (Jones and Bergen, 2024) and near-chance for prior models, and a subsequent three-party test reached 73% for GPT-4.5, above the human baseline (Jones and



Bergen, 2025a). These are ordinal claims established by blind human judgment rather than self-reported capability scores, and they place the crossing of near-human indistinguishability in the November 2023–March 2024 window (Figure A1). We acknowledge two caveats: no benchmark measures indistinguishability of governance-forum prose specifically, so the Arena and Turing tests are the nearest available instruments; and the highest Turing pass rates require prompting the model to adopt a humanlike persona.

The benchmark evidence described here does not independently locate the indistinguishability crossing in the November 2023–March 2024 window. The Turing-test observations involve different models and protocols, while Chatbot Arena measures relative output preference rather than human classification of governance prose. Together they support a broad rise in humanlike capability, but the narrow timing remains suggestive rather than established.

#60 Contradiction between subsidy claim and core mechanism

subsidy attaches to observable text; once capable AI makes text free to produce ($c_p^{AI} \approx 0$), each subsidized zero-content rationale pushes the delegate's expected contamination π toward the tipping point π^ , and past π^* the collapse is discrete and total. A program that pays for the observable proxy of engagement, deployed once that proxy has become costless to counterfeit, can therefore finance the destruction of the very forum it exists to support*



The subsidy implication is overbroad as applied to Arbitrum's observed collapse. Subsidies can raise expected contamination π without implying mechanical dilution by realized AI volume, so the general policy mechanism is coherent; however, Arbitrum's DIP began after the paper's estimated November 2023 break and therefore cannot have financed that particular collapse.

#61 Misinterpretation of log10 regression coefficient



Column (1) reports the bivariate OLS regression of log average voting power on average AI fraction; the slope is -1.84 ($t = -3.47, p = 0.001$), implying that a 10-percentage-point increase in average AI fraction is associated with a 17% decline in voting power.

The 17% interpretation is inconsistent with Figure 9's base-10 scale. If the reported -1.84 coefficient comes from a $\log_{10}$ regression, a 0.10 increase in AI fraction implies $10^{-0.184} - 1 \approx -34.5\%$, not 17%. The plotted line also appears inconsistent with Table 4's reported intercept, further suggesting that the figure and regression output may not use the same transformation or sample.

#62 Threshold robustness overstates attenuation

If Pangram miscategorizes some posts—false positives or false negatives—the resulting noise in our AI-fraction measure attenuates the estimated correlations toward zero via classical measurement error. Any such bias therefore works against finding a significant structural break, making our reported $r_{\text{pre}} = -0.33$ and Chow $F = 8.38$ conservative estimates of the true relationship.

The claim that Pangram misclassification necessarily attenuates the correlations and makes the Chow statistic conservative is not established. Because human-post counts aggregate thresholded classifications, error rates may vary with thread size, writing style, AI prevalence, and period; such differential misclassification can bias either regime's slope—and hence the estimated break—in either direction. The threshold exercise establishes cutoff insensitivity, not the direction of bias from classifications that are wrong at every tested cutoff.

#63 Robustness claim conflicts with later break tests

The change in the correlation coefficient, $\Delta r = +0.23$, is confirmed by a Chow test that rejects parameter stability at $F = 8.38$ ($p = 0.0004$). We use Claude 3 (March 2024) as our primary break date, which provides a clean before-and-after contrast. All substantive conclusions hold at any candidate break date from November 2023 onward (Table 3).

The claim that all substantive conclusions hold at every candidate breakpoint from November 2023 onward conflicts with Table 3. The o1 and DeepSeek R1 splits do not reject parameter stability, and the DeepSeek split has a more negative post-break correlation than pre-break correlation. The supported robustness window is therefore narrower than the sentence states.

#64 Generic DAO definition overstates automation

*A decentralized autonomous organization (DAO) is an entity whose governance rules are encoded as smart contracts on a blockchain and executed automatically without the discretion of any central authority. The defining features are: (i) membership is determined by ownership of a governance token, typically a fungible ERC-20 asset (a standardized transferable digital token on the Ethereum blockchain) tradeable on secondary markets; (ii) decisions are made by on-chain votes in which voting power is proportional to token holdings; and (iii) the outcomes of votes are implemented deterministically by the protocol, not by human managers who retain the discretion to deviate. DAOs differ from traditional firms in that there is no board of directors, no CEO, and no legal person behind the organization—only code and the community of token holders who govern it.*¹

The definition presents an idealized, fully on-chain DAO as universal, but Arbitrum's described architecture is hybrid: forum and Snapshot stages are off-chain, delegation mediates token voting, Foundation and council actors administer parts of governance, and only a later Tally stage is binding. On-chain proportional voting, deterministic implementation without human discretion, and the absence of any involved legal person are therefore not established as defining features of the institution studied.

#65 Mischaracterization of influential outliers

Two observations exceed the conventional high-influence threshold of $|D|/n=0.138$

: a July 2023 treasury – spend proposal for the Camelot ecosystem hub (a high – posts / high – margin point that attenuates the $F(F=5.62, \text{permutation } p=0.010)$).



Leave-one-out analysis. Figure 10, Panel B reports the distribution of r_{pre} computed across all 29 leave-one-out subsamples. Values range from -0.14 to -0.42 (mean = -0.33 , s.d. = 0.042), with 97% of removals yielding $r < -0.20$. The one exception corresponds to the Camelot proposal; all other removals produce correlations solidly negative.

The influential observations are described in the wrong direction. The high-post-count observation in Figure 10 has a low margin and steepens the negative relationship; consistently, the text says that removing the Camelot proposal moves the correlation from -0.33 to approximately -0.14 . The low-post-count, low-margin observation is the one that attenuates the negative slope. The proposal identities and their respective influence should be reconciled with the plotted diagnostics.

#66 Observational equivalence claim is too broad

Text-based financial signaling. Textual analysis in finance has documented that the content of corporate communications carries information: Li (2008) shows that more readable annual reports forecast better earnings; Loughran and McDonald (2011) construct a finance-specific word list that predicts stock market



reactions; Cookson and Niessner (2020) find that disagreement among social-media investors predicts subsequent trading volume. These papers treat authorship as unambiguous. Our paper is, to our knowledge, the first to study what happens when the link between authorship and authentic opinion is broken by AI, and to show that this break destroys the informational content of text-based signals even when the text is observationally unchanged.

The claim that the text is “observationally unchanged” is stronger than the evidence supports. Pangram classifies posts from textual features, and the paper reports systematic differences in vocabulary, length, and generic agreement language. The mechanism supported by the analysis is costly or unreliable post-level verification by ordinary delegates, not literal observational equivalence of human and AI text.

#67 Inconsistent delegate count between analyses

Figure 9 presents the cross-sectional relationship between delegate AI adoption and delegated voting power. Each observation is one of the 51 delegates for whom we observe both Pangram AI fraction (averaged across all their forum posts) and average voting power



The delegate counts are internally inconsistent: the cross-section and data-construction discussion identify 51 delegates linked to both Pangram scores and Snapshot voting records, whereas Table 5 reports 54 delegates using those same data types. The panel could legitimately use different eligibility rules, but those rules or the source of the three additional linked delegates are not stated.

#68 DIP incentive is scaled by delegated, not owned, ARB

An institutional microfoundation. Arbitrum supplies a concrete pecuniary motive for this pattern. The Delegate Incentive Program (DIP), administered by SEEDGov, pays eligible delegates up to 5,000 ARB per month for voting and publicly posting a rationale for each vote. The subsidy attaches to observable text, and its value relative to a delegate's stake is largest precisely for lower-power delegates: for a delegate holding tens of thousands of ARB in delegated power, 5,000 ARB per month is a material return to producing rationale posts at minimum cost, whereas for a delegate holding millions it is negligible. DIP therefore gives the cross-sectional pattern an institutional rationale—it rewards exactly the low-power tail of the 27× voting-power gap for generating posted text—and AI lowers the cost of manufacturing the subsidized observable (a plausible-sounding rationale) without the underlying analysis.

The DIP argument conflates delegated voting power with the delegate's own economic stake. Because delegation transfers voting authority without transferring ownership, the ratio of a 5,000 ARB payment to delegated ARB does not establish that the payment is financially more material to low-power delegates. DIP plausibly creates an incentive to minimize the cost of producing required rationales, but the reported 27× delegation gap alone does not show that this incentive is stronger for delegates with less delegated power.

#69 Figure A2 caption describes a missing taxonomy plot

Figure A1: Voter Turnout and Quorum Adequacy Arbitrum DAO – Snapshot Votes, 2023–2026 (% of delegated/votable supply; quorum thresholds marked)

Figure A2: Economic Taxonomy of Arbitrum DAO Proposals. Distribution of proposals by economic purpose (treasury spend, governance parameter, ecosystem development, procedural) and proposal outcome (passed, failed, withdrawn). Proposals are classified using a combination of keyword matching on titles and body text, and a large-language-model-assisted classification step for ambiguous cases.

The Figure A2 caption does not correspond to the adjacent graphic. The image is a turnout-over-time plot split between constitutional and non-constitutional tracks, while the caption describes a distribution by proposal category and outcome. The promised taxonomy and outcome evidence is therefore not displayed in this appendix entry.

#70 Contradictory use of “notional numbers”

These are not notional numbers: DAO treasuries consist primarily of the protocol's own governance token plus stablecoins and other liquid assets, held in smart contracts that can be disbursed only by a governance vote.

The statement that the treasury values are “not notional” conflates the existence of the tokens with the realizability of their mark-to-market USD value. Because much of the treasury is the DAO's native token, the headline \$3.5 billion valuation can be highly sensitive to price and market depth and generally cannot be assumed realizable through liquidation, although the tokens do represent genuine allocative assets.

#71 Mathematical contradiction in the calibration threshold

The threshold condition then gives $c_{\text{read}} / I_H = 1 - \pi^ \leq 0.92$: reading costs absorb at most 92% of the informational value of a perfectly reliable post. This is economically plausible in a setting where (i) each individual post contributes modestly to aggregate signal (the forum aggregates across hundreds of posts per proposal, so marginal informational value per post is small), (ii) careful reading of a governance post requires domain-specific knowledge and real attention, and (iii) a delegate cannot determine a post's type without first reading it, so expected value must exceed cost before type is revealed. Even a 5% probability of encountering a fully uninformative AI post is sufficient to erode the thin net return to deliberate forum reading.*

The characterization of the net return as “thin” is not established by $c_{\text{read}} / I_H \leq 0.92$: this upper bound implies only that the reliable-post surplus share $[0, \pi^*] = 1 - c_{\text{read}} / I_H$ is at least 8%, not that it is close to 8%. A 5% contamination probability does reduce expected reading value, but it is not shown to cross the exit threshold.

#72 Figure 9 leaves the regression sample unclear

Figure 9: Delegate AI Adoption and Voting Power: Cross-Section. Each point represents one delegate ($n = 51$). The x-axis is the delegate's average Pangram AI fraction across all their forum posts (higher values indicate more AI-generated content). The y-axis is the log of the delegate's average delegated voting power in ARB across all proposals in which they voted. Marker size is proportional to total forum post count. The fitted OLS line has slope -1.84 ($t = -3.47, p = 0.001$); the gray band is the 95% confidence interval. Selected delegates are labeled. High-AI delegates hold systematically lower voting power: those with AI fraction above 20% average 71,000 ARB versus 1.9 million ARB for low-AI delegates.

The Figure 9 sample appears internally inconsistent. Its annotation reports $r = -0.439$ for 51 “individual delegates,” while the plot separately includes institutional delegates; Table 4 also has 51 observations and identifies an institutional-delegate coefficient. It is therefore unclear whether the correlation, fitted line, group means, and regressions use all 51 linked delegates or an individual-only subsample.

#73 Figure 2 conflicts with the stated 33% minimum

The distribution of voting power is highly concentrated. The Nakamoto coefficient—the minimum number of independent actors required to collude and control a majority of votes—stands as low as 4 delegates at the 51% threshold and 3 at the 33% blocking-minority threshold in early windows (Figure 2).

Figure 2: Evolution of Voting Power Concentration. Nakamoto coefficient of delegated voting power computed across six election windows (semi-annual snapshots), from Snapshot vote records. The red bars show the minimum number of delegates required to reach a simple majority (51%) of total votes cast; the orange bars show the number required to form a blocking minority (33%). Lower values indicate more concentrated power.

Figure 2 labels the September 2023 Nakamoto coefficient at the 33% blocking-minority threshold as 1, whereas the text says the early-window minimum was 3. Because both passages appear to describe the same statistic and sample windows, one of these values is incorrect.

#74 Inconsistent net social benefit figure in abstract

Calibrating to the data, the forum generated net social benefits equivalent to saving at least 858 words of research effort per delegate per proposal—all destroyed at the tipping point.

The abstract conflates net reading value with net social surplus. The 858-word figure is net of a delegate's reading cost but not of the 10,480-word contributor cost per proposal; at $K = 62$, the paper's social-surplus formula implies approximately 689 words per delegate.

#75 Unexplained discrepancy in TF-IDF corpus size

We characterize AI post content directly using TF-IDF analysis of all 1,313 AI-classified posts against the 13,458 human-authored posts in the corpus.

The TF-IDF sample contains 1,313 AI-classified and 13,458 human-authored posts, totaling 14,771, whereas Section 3.2 reports 15,891 scored posts. Given the paper's binary threshold classification, 1,120 scored posts are unaccounted for in the content analysis, and no additional TF-IDF eligibility rule is reported.

#76 Ambiguity between QLR date and Chow test date

The QLR test identifies a structural break at November 25, 2023 ($\sup F = 17.19$, $5\%CV = 8.85$), nineteen days after GPT-4 Turbo was released. The Chow test at that date produces the largest F -statistic in the battery ($F = 15.05$, Table 3).

The date associated with $F = 15.05$ is unclear. The sentence appears to assign it to the November 25 QLR break date, while Table 3 assigns it to the November 6 GPT-4 Turbo release date; the QLR statistic at November 25 is separately reported as 17.19. These statistics and their corresponding split dates should be distinguished precisely.

#77 Reversed reading threshold in Section 4.1 and 7.1

Under the unverifiability mechanism, the relevant shock is to the verifiability of post authenticity. Let π denote a delegate's posterior probability that an observed post is AI-generated. When π is near zero, forum posts are informative signals. When a sufficiently capable LLM is publicly released, AI-generated posts become indistinguishable from human-authored contributions on all observable features. Bayesian updating on observables can no longer resolve π below a positive lower bound $\pi > 0$. If $\pi \cdot IH < c_{\text{read}}$ (the informational gain from a post falls below the cost of reading), delegates rationally discount the forum. The credible possibility of unverifiable low-quality entries is sufficient to destroy the information value of high-quality entries, even when good entries predominate.

The reading condition is reversed here. Because π is the probability that a post is AI-generated and AI posts are assumed uninformative, the expected informational benefit is $(1 - \pi)IH$, not πIH . The stated condition has the opposite comparative static from the unverifiability mechanism formalized later in the paper. This error also appears in Section 7.1, where it states delegates exit when $\pi \cdot IH < c_{\text{read}}$. Given the paper's definition of expected reading value, delegates exit when $(1 - \pi)IH < c_{\text{read}}$.

#78 Mathematical contradiction in voting power concentration

By the end of our sample period (March 2026), the top 10 delegates by voting power collectively controlled approximately 30% of all votes cast in contested proposals, with the largest single delegate holding over 1% of total circulating ARB supply in delegated power.

There seems to be an issue with the denominators: the 30% figure concerns votes cast in contested proposals, while Figure 2 alternately describes delegated voting power and total votes cast. The figures are not mathematically inconsistent if they refer to different populations, but their relationship cannot be interpreted reliably unless the denominator, proposal sample, and aggregation method for each statistic are distinguished.

#79 Vestigial text misstates the paper's central empirical challenge

The welfare effects of vote markets depend on whether votes are cast on the basis of private information or private preferences (Riker and Ordeshook, 1968; Casella et al., 2012); if buyers have superior information, vote markets aggregate it more efficiently than one-token-one-vote; if buyers simply have more money, vote markets allow wealth to override collective judgment. Distinguishing these cases empirically is the central challenge in interpreting our results.

The claim that distinguishing informational from wealth-driven vote buying is “the central challenge in interpreting our results” overstates the role of the vote-rental question. The paper does not empirically distinguish those cases; its central interpretation instead concerns whether the forum-signal break reflects AI-related unverifiability rather than realized AI volume or concurrent confounders.

#80 Figure 1 does not show the stated post-volume peak

Magnitude: forum volume ranges from a low of roughly 200 posts per month in the early sample to a peak of over 1,500 in mid-2024, a sevenfold range. Timing: the post-count series leads the price series at several turning points, particularly around major grant program votes in late 2023 and the ecosystem incentive cluster in early 2024. This leading pattern motivates the question of whether forum activity aggregates forward-looking information—which our main analysis answers in the affirmative for the pre-AI period.

Figure 1: Forum Participation and Token-Market Co-movement. Monthly Arbitrum governance forum post count (blue bars, left axis), ARB/USD month-end closing price (dark line, right axis), and monthly ARB trading volume in USD billions (pale blue shaded area, right axis).

Figure 1 appears inconsistent with the stated peak of more than 1,500 monthly posts in mid-2024: its left axis tops out at 1,400, and the tallest displayed bars are approximately 1,200–1,250. Consequently, the stated endpoints of roughly 200 and more than 1,500, and hence the claimed sevenfold range based on those endpoints, are not supported by the displayed series.

#81 Inconsistent delegate count in the Introduction

Using a dataset linking forum usernames to Snapshot voting records for 60 delegates, we document a robust negative cross-sectional relationship between a delegate's average AI content fraction and their average delegated voting power: $r = -0.44$ ($p = 0.001$, $n = 51$ delegates).

The 60-delegate linked dataset is not reconciled with Section 3.5's 62 wallet-username matches, 52 delegates with at least 10 posts, and 51 linked observations used in the cross-section. These may be nested samples, but the document does not explain how the count of 60 is obtained, leaving the sample attrition unclear.

#82 Antweiler and Frank (2004) contradicts the claim about predicting future returns BETA

Retail investors post on Reddit and StockTwits, and a large empirical literature has established that this crowd activity aggregates real information about future returns (Antweiler and Frank, 2004; Chen et al., 2014).

The cited paper by Antweiler and Frank (2004) contradicts the claim. While they study internet stock message boards, they explicitly conclude that this activity does *not* meaningfully predict future stock returns, although it does help predict volatility and trading volume.

#83 Citation identifier resolves to a different work BETA

On the Massive Multitask Language Understanding benchmark (MMLU; Hendrycks et al., 2021), which tests breadth of knowledge across 57 academic and professional domains, GPT-4 (March 2023) scored 86.4% versus 70.0% for GPT-3.5—a step change that closed most of the remaining gap to expert human performance (estimated at 89.8%).

The citation identifier provided (arXiv:2008.02275) and the title “Aligning AI with shared human values” point to a paper introducing the ETHICS dataset, not the MMLU benchmark discussed in the text. The MMLU benchmark was published in a different paper by Hendrycks et al. titled “Measuring Massive Multitask Language Understanding” (arXiv:2009.03300). Because the identifier points to a different work, the citation is unverified as cited.

#84 GPT-4 technical report does not cover GPT-4 Turbo or this specific finding BETA

Most relevant for governance prose, evaluations of long-form argumentative and analytical writing—the genre of DAO forum posts—show a discrete jump precisely at GPT-4 Turbo, with evaluators unable to reliably distinguish outputs from expert human writers at threshold quality settings (OpenAI, 2023).

The cited GPT-4 technical report (arXiv:2303.08774) was published in March 2023, months before the release of GPT-4 Turbo. It does not contain an evaluation of long-form argumentative writing showing that evaluators are unable to reliably distinguish GPT-4 Turbo outputs from expert human writers. This finding is either misattributed or absent from the cited work.

#85 Lopez-Lira and Tang (2023) studies news-based stock forecasts, not earnings calls BETA

Recent work documents that AI affects labor market outcomes (Brynjolfsson et al., 2023; Eisefeldt et al., 2023), financial analysis quality (Cao et al., 2024; Kim et al., 2024), and earnings call dynamics (Lopez-Lira and Tang, 2023).

The identifiable 2023 paper by Alejandro Lopez-Lira and Yuehua Tang is Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models, not ChatGPT and the Labor Market. It uses news headlines to test whether ChatGPT sentiment scores predict subsequent daily stock returns. It does not study earnings-call dynamics, labor-market outcomes, or workers using AI as a productivity assistant. The citation therefore points to a different subject and does not support the proposition attributed to it. ([arxiv.org](https://arxiv.org/abs/2303.08774))

#86 Frankenreiter & Nyarko (2022) citation could not be located BETA

For an overview of DAOs and their relationship to traditional corporate law, see Frankenreiter and Nyarko (2022); for the economics of decentralized finance, see Harvey et al. (2021) and Jensen et al. (2021).

I could not locate any paper titled 'The promise and limits of private ordering in blockchain-based contracts' by Frankenreiter and Nyarko. Extensive searches for this exact title, as well as searches for any work by these authors addressing DAOs and traditional corporate law, returned no matches. The authors did co-author a 2022 book chapter titled 'Natural language processing in legal tech', but it does not cover blockchain or corporate law. This citation appears to be fabricated (a hallucination) and should be replaced with a valid source.

#87 Kamenica and Gentzkow (2011) explicitly does not use soft information BETA

Relative to this literature, and to models of biased communication with soft information (Che and Kartik, 2009; Kamenica and Gentzkow, 2011), our contribution is on the demand side of persuasion—the principal is not a passive processor of whatever advocates submit but pays c_T per post and rationally chooses whether to listen at all, ceasing entirely past π^ .*

The submission categorizes Kamenica and Gentzkow (2011) as a model of "biased communication with soft information." This is a material mischaracterization of the literature. Kamenica and Gentzkow's seminal "Bayesian persuasion" model explicitly relies on the sender's ability to commit to an information structure that generates verifiable, hard signals, and is routinely contrasted in the literature with soft-information (cheap talk) models.

#88 Luskin et al. (2002) studies an in-person event, not online forums BETA

A growing literature studies whether online forums support genuine deliberation (Luskin et al., 2002; Wright, 2012).

The submission cites Luskin et al. (2002) as studying "whether online forums support genuine deliberation." However, the cited paper analyzes the first British Deliberative Poll, which was conducted entirely offline as a weekend face-to-face gathering in Manchester in 1994. It contains no study of online forums.

#89 Jabarian & Imas (2025) contradicts the claim that most commercial detectors fail BETA

Not all AI detectors perform equally well: Imas and Jabarian (2024) audit a range of commercial AI-text detectors and find that most fail to reliably distinguish AI-generated from human-written text, while Pangram achieves materially higher accuracy.

The cited study—actually titled "Artificial Writing and Automated Detection" (Jabarian & Imas, 2025)—contradicts the claim that most commercial AI detectors fail. The authors evaluated three commercial tools and one open-source tool, explicitly concluding that "most commercial AI detectors perform remarkably well" and generally keep false positive rates below 1%. While the study confirms that Pangram achieved the highest accuracy and was uniquely robust on very short or altered texts, it attributes widespread failure entirely to open-source alternatives (which misclassified up to 78% of human text), not the commercial tools as the submission claims.

#90 Verba et al. (1995) rejects strict rational choice models of participation BETA

When participation is costly and agents engage only when their expected benefit exceeds the cost (Downs, 1957; Verba et al., 1995), equilibrium forum activity is increasing in the degree of community division: proposals about which opinion is roughly split ($\theta_i = 0.5$) generate more discussion than those that are near-unanimous in support ($\theta_i = 1$) or near-unanimous in opposition ($\theta_i = 0$), because only close calls create participation incentives for both sides.

Verba et al. (1995) is materially misrepresented here. Their book, *Voice and Equality*, developed the Civic Voluntarism Model explicitly as a critique of and alternative to narrow rational choice theories (like that of Downs). They point out that a strict expected-benefit-exceeds-cost calculus cannot adequately explain why people participate in politics, and instead emphasize resources (time, money, civic skills) and recruitment networks. Citing them in support of the strict condition that agents participate *only* when expected benefits exceed costs contradicts the core thesis of their work.

#91 Andrews (1993) critical value of 8.85 is for 1 parameter, not 2 BETA

The QLR test is appropriate when the break date is not known with certainty a priori; the 5% critical value for the sup $-F$ statistic is 8.85 (from Andrews 1993, Table 1, for $k = 2$ parameters).

Andrews (1993) is misrepresented regarding the specific critical value. Table 1 of the paper provides the critical value of 8.85 for a test of a break in exactly *one* parameter ($k = 1$) with 15% trimming at the 5% significance level. For $k = 2$ parameters, the 5% critical values are different (the sup-Wald critical value is 11.79, corresponding to a sup-F critical value of 5.89). Attributing the 8.85 threshold to a 2-parameter test is incorrect and applies the wrong threshold for statistical significance.

#92 Statistic about trained evaluators not found in Anthropic (2024) BETA

Prior models (GPT-3.5, Claude 1, Claude 2) produced outputs that trained evaluators could identify with roughly 70-75% accuracy; GPT-4 Turbo and Claude 3 reduced this to near chance (Anthropic, 2024).

The cited Claude 3 model family report (Anthropic, 2024) does not contain the specific finding that prior models were identified with 70-75% accuracy by trained evaluators while Claude 3 reduced this to near chance. While the report discusses extensive model evaluations, this specific statistic regarding human ability to identify AI-generated outputs is absent.

#93 Che and Kartik (2009) is a model of hard information, not soft information BETA

Relative to this literature, and to models of biased communication with soft information (Che and Kartik, 2009; Kamenica and Gentzkow, 2011), our contribution is on the demand side of persuasion—the principal is not a passive processor of whatever advocates submit but pays c_T per post and rationally chooses whether to listen at all, ceasing entirely past π^ .*

The submission mistakenly classifies Che and Kartik (2009) as a model of “biased communication with soft information.” The cited paper relies on verifiable (hard) information, explicitly noting that while advice is strategic, the adviser “cannot falsify information.” Soft information typically refers to unverifiable cheap talk, which this paper is not modeling.

#94 Riker and Ordeshook (1968) does not discuss vote markets or information aggregation BETA

The welfare effects of vote markets depend on whether votes are cast on the basis of private information or private preferences (Riker and Ordeshook, 1968; Casella et al., 2012); if buyers have superior information, vote markets aggregate it more efficiently than one-token-one-vote; if buyers simply have more money, vote markets allow wealth to override collective judgment.

The cited work, "A Theory of the Calculus of Voting" (1968), introduces a foundational model of voter turnout based on expected utility (explaining why people vote despite the costs). It does not discuss vote markets, nor does it analyze how voting based on private information versus private preferences affects the welfare outcomes of voting. The submission incorrectly attributes the distinction between private information and private preferences in vote markets to this paper.

#95 DeFiLlama treasury data misattributed BETA

Source: DeFiLlama, June 2024.

DeFiLlama data from mid-2024 actually shows that Uniswap held the largest DAO treasury at roughly \$3.5 billion USD. While Arbitrum's treasury contained roughly 3.5 billion of its native ARB tokens, their actual USD value at the time was around \$1.4–\$1.8 billion, making it the second-largest treasury. The submission appears to confuse Arbitrum's token count (3.5 billion ARB) with its USD value, incorrectly claiming it held the single largest treasury at \$3.5 billion.